# QUEZON CITY UNIVERSITY

**COLLEGE OF COMPUTER STUDIES**

**MS101 – DISCRETE MATHEMATICS**

**ACTIVITY 1**

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Instruction: Solve for the following. Show your solution. (15 pts.)

1. **Roll of 1 die**

   What is the probability that the outcome is 3?

   Total outcomes when rolling a die = 6 (1, 2, 3, 4, 5, 6)

   Favorable outcome = 1 (the number 3)

   **Answer:** $\frac{1}{6}$

2. **Roll of 2 dice**

   What is the probability that the outcome is 6?
   Possible pairs (die1, die2) that sum to 6: (1,5), (2,4), (3,3), (4,2), (5,1)

   Favorable outcomes = 5
   Total possible outcomes = 6×6=366 \times 6 = 366×6=36

   **Answer**: $\frac{5}{36}$

3. **You are flipping a coin 3 times, the first flip is a head.**
   Given this, what is the probability that the 3 flips produce an even number of heads?

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**Step 1:**
Since the first flip is fixed as **Head (H)**, the first head is already guaranteed.
Now we only need to look at the remaining **2 flips**.

**Step 2:**
We need a total *even number* of heads overall.
We already have **1 head**, so we need **1 more head** (to make 2 heads total, which is even).

**Step 3:**
The last two flips can be:
HH → total 3 heads (odd)
HT → total 2 heads (even) ✅
TH → total 2 heads (even) ✅
TT → total 1 head (odd)

Favorable outcomes = 2 (HT, TH)

Total possible = 4 (HH, HT, TH, TT)

**Answer:** $\frac{1}{2}$